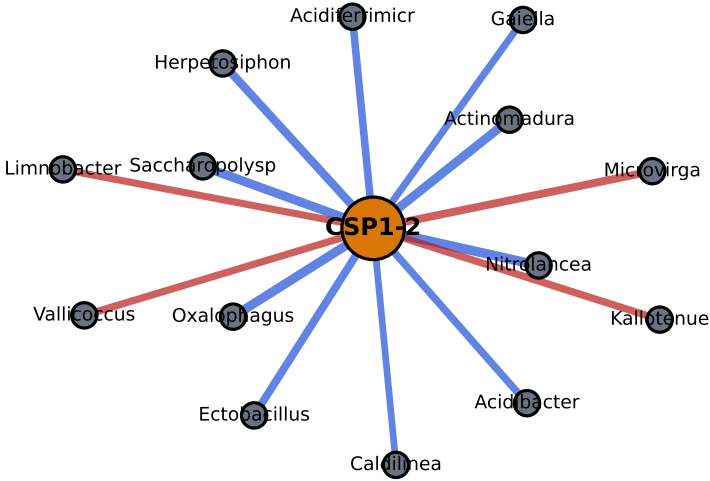
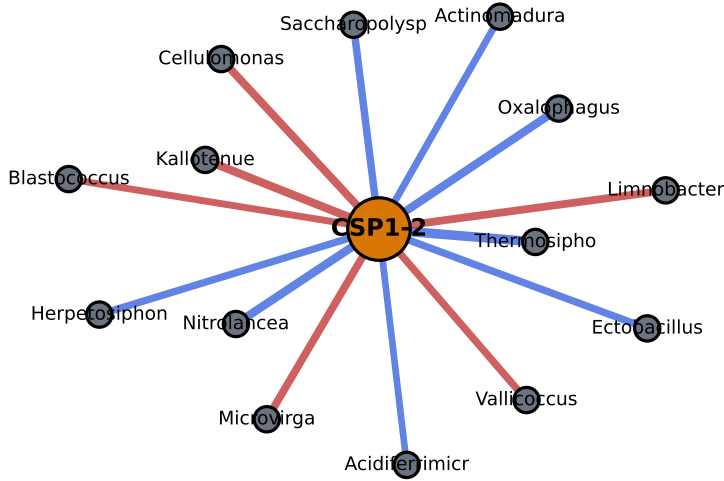


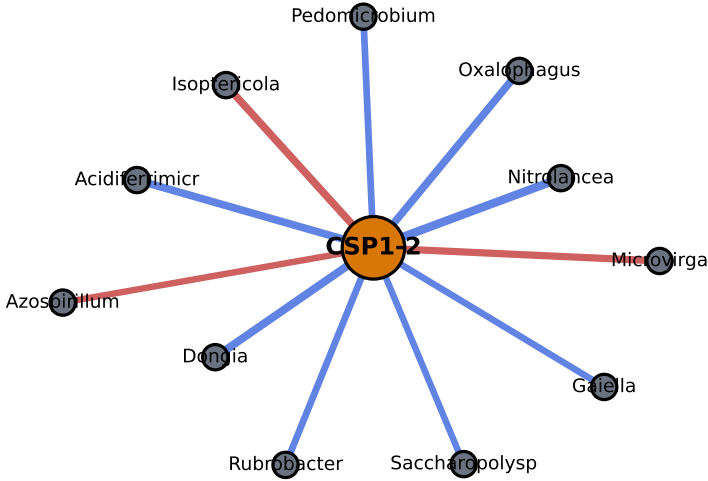
(a) Surface — CSP1-2 ego network (14 co-occurring genera)



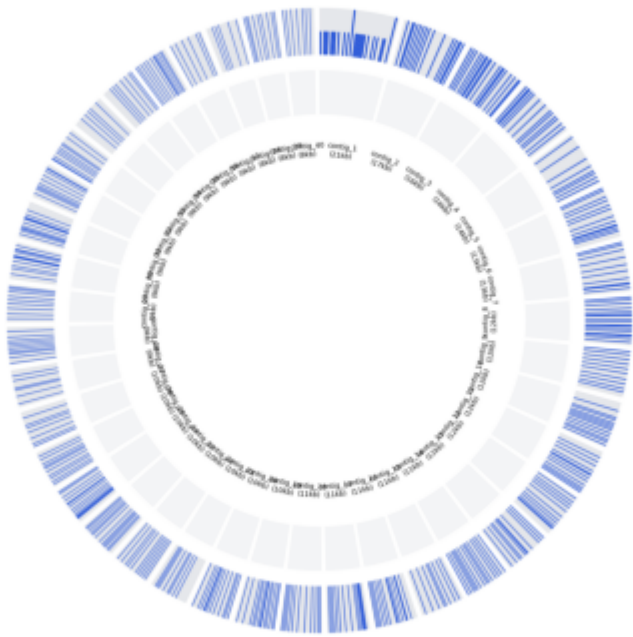
(b) Deep — CSP1-2 ego network (14 co-occurring genera)



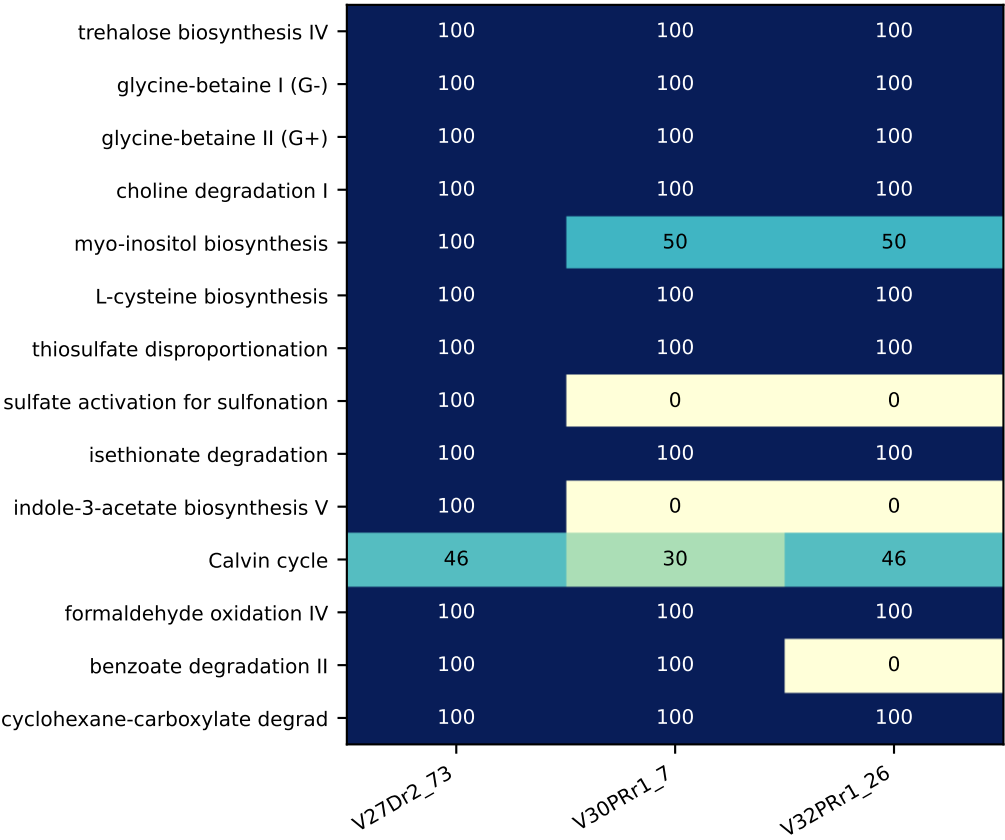
(c) Rhizosphere — CSP1-2 ego network (14 co-occurring genera)



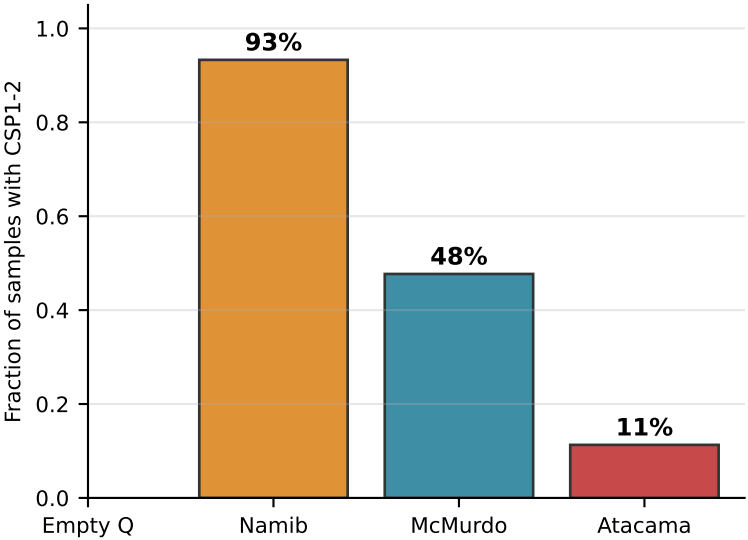
(d) Reference MAG V27Dr2_SemiBin_73 (2.66 Mb, GC 43.1%) — top 40 contigs
outer ring: CDS density (blue). 0 CRISPR arrays. BGCs: NRPS-like, RRE, T3PKS (antiSMASH).



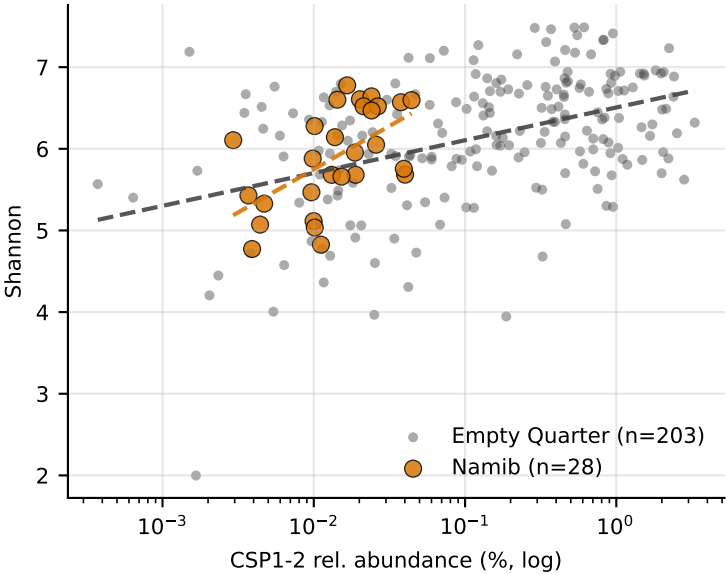
(e) Signature pathways across MAGs
(% completeness, gapseq default)



(f) Cross-desert prevalence



(g) CSP1-2 abundance vs Shannon (EQ + Namib)
EQ: $\rho=+0.37$, $p=4.4e-08$ | Namib: $\rho=+0.58$, $p=1.1e-03$



(h) Salinity vs Shannon (EQ + Atacama)
EQ (XRF S): $\rho=-0.34$, $p=1.4e-06$ | Atacama (EC): $\rho=-0.38$, $p=1.2e-02$

